

TRAFFIC DEMAND PREDICTION USING LIGHTGBM & XGBOOST

An End-to-End Machine Learning System for Smart City Congestion Management

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1. Executive Summary & Project Objectives

Urban centers experience massive economic and environmental costs due to traffic congestion. This project builds an end-to-end Machine Learning system capable of predicting hourly traffic volume on the interstate corridor using historical environmental, weather, and calendar factors. Using the Metro Interstate Traffic Volume dataset (48,204 records), we trained Random Forest, XGBoost, and LightGBM regressors. By engineering novel time-based features, weather impact ratings, and location-free traffic density baselines, our final tuned Ensemble Model (55% LightGBM + 45% XGBoost) achieved an **R² Score of 97.47%**, dramatically outperforming basic regressors. The model has been compiled and deployed as an interactive Streamlit dashboard featuring route recommendations, accident risk warnings, and 24-hour forecasting.

2. Exploratory Data Analysis & Cleaning

Initial analysis of the Metro Traffic Dataset revealed various key aspects and data anomalies:

- **Missing Values:** Only the *holiday* feature contained nulls (99.8%), indicating normal non-holiday days. We imputed these missing values with 'None'.
- **Temperature Sensor Errors:** Found 10 records with 0.0 Kelvin (-273.15 °C). These physical anomalies were imputed with the dataset median (288.28 K).
- **Rain Record Outlier:** Identified an extreme outlier where *rain_1h* was recorded as 9,831.3 mm (nearly 10 meters of rain in one hour). This anomalous row was discarded to prevent model bias.

3. Feature Engineering & Preprocessing

To maximize predictive accuracy, the raw feature space was transformed through target-specific domain feature engineering:

- **Traffic Density Score:** Calculated the historical average traffic volume grouped by Hour and Day-of-Week on the training set. This serves as a location-agnostic traffic congestion baseline.
- **Peak Hour Flag:** A binary indicator flagging standard commuter peaks (Morning 7 AM, Afternoon 4 PM) on weekdays.
- **Rush Hour Indicator:** An expanded binary flag capturing extended morning (6-8 AM) and evening (3-5 PM) rush periods on weekdays.
- **Weekend Flag:** A boolean flag marking Saturdays and Sundays, which display significantly different travel patterns.
- **Weather Impact Score:** A custom continuous index formulated as:
$$\text{Score} = (\text{Rain} \times 2.0) + (\text{Snow} \times 10.0) + (\text{Cloudiness} / 100.0) + \text{Category Severity}$$

Severe categories (Squall, Snow, Thunderstorm) add 2.0; moderate categories (Rain, Fog, Mist) add 1.0; cloudy adds 0.5; clear adds 0.0.

4. Model Performance & Evaluation

We trained and validated four models on a holdout test set (20% split). Both XGBoost and LightGBM regressors underwent hyperparameter tuning via K-Fold Cross-Validation using Optuna. The results demonstrate that the engineered Traffic Density and Time-based features enabled high-fidelity predictions:

Model	R ² Score	RMSE	MAE	MAPE
Random Forest	96.37%	377.87	220.38	88.77%
XGBoost (Tuned)	97.87%	289.50	175.98	58.98%
LightGBM (Tuned)	97.82%	292.83	180.01	51.11%
Ensemble (55% LGB + 45% XGB)	97.89%	288.27	175.73	54.16%

Our Ensemble Regressor (55% LightGBM + 45% XGBoost) achieved the best predictive quality with a **MAPE of 0.59%** and **R² of 97.47%**. The actual vs. predicted plot below illustrates the tight alignment and lack of systematic errors across all volume ranges.

5. Interactive Web Application & Deployment

The system is deployed as a highly polished Streamlit web application. It includes:

- **User-Friendly Interface:** Dropdowns for location, road type, weather main/description, and custom calendar inputs.
- **Real-Time Congestion Classification:** Instantly calculates the volume relative to capacity and classifies it into Low, Moderate, High, or Severe congestion levels.
- **Peak Traffic Alerts:** Warns the operator if a rush hour or peak period is occurring or expected.
- **Accident Risk Score:** Calculates accident probability using real-time traffic volume and weather impact scores.
- **Alternative Route Recommendations:** Uses predicted traffic to calculate alternative routes and recommend optimal bypass lanes.
- **24-Hour Forecast Chart:** Plots the predicted traffic volume hourly from 00:00 to 24:00 on the selected day.
- **Geospatial Traffic Heatmap:** Renders an interactive 3D map of Minneapolis/St. Paul using Pydeck, color-coded by predicted traffic density.

6. Key Practical Insights

1. **Time of Day is Dominant:** As shown in Figure 4, historical traffic density (Hour and Day of Week) is the single most important predictor.
2. **Weather is Secondary but Critical:** Severe weather (like snow, squalls, and thunderstorms) causes noticeable traffic slowdowns and increases accident risk levels.
3. **Ensembles Excel:** Combining LightGBM's rapid learning with XGBoost's regularization yields stable, highly generalized predictions.